Arthritis Foundation

June 16, 2025

The Honorable Mehmet Oz
Centers for Medicare and Medicaid Services
U.S. Department of Health and Human Services
Room 445-G-Hubert H. Humphrey Building
200 Independence Avenue SW
Washington, DC 20201

RE: [CMS–0042–NC] RIN 0938–AV68 Request for Information; Health Technology Ecosystem

Dear Administrator Oz:

On behalf of the nearly 60 million American adults and 300,000 children living with arthritis, the Arthritis Foundation appreciates the opportunity to comment on the Request for Information (RFI) on the health technology ecosystem. Access to health technology to manage the administrative care and treatment of arthritis is critically important to fostering a truly patient-centered system of care. CMS rightly notes that better implementation and use of digital tools can bring much-needed transparency to patients and streamline the coordination of their care. Patients continuously report care coordination as a major challenge in surveys and focus groups and we applaud CMS for prioritizing the health technology ecosystem and seeking information from patients to inform its work.

Below please find our comments on Sections B and F. We surveyed a small cohort of 37 patient advocates for their feedback on the patient-facing questions, and have supplemented our comments with other Arthritis Foundation data sets and policy positions.

Section B.1 – Patient and Caregiver Feedback

1. Patient Needs

PC–1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

> **a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?**
> **b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.**

We asked patients in our survey to indicate which functions they would want an app to perform. Their answers in order of most checked item were: see test results (34

respondents); read provider notes (32); check prescription status (3); make appointments (30); get reminders (25); help file appeals (20); track PROs (1); ability to message nurse or billing department (1); manage copay cards (1); track symptoms (1).

We then asked patients what functions would be most important for digital tools to help manage their care. The results were: coordinate medication information across all my doctors (27 respondents); keep track of symptoms (4); explain lab results (3); find a good doctor (2); manage appeals (1).

Finally, we asked what they would ask a personal assistant to help with in navigating their care. Their answers were: coordinate medical information across all my doctors (32 respondents); keep track of symptoms (20); find a good doctor (18); explain lab results (16); remind you to take your medicines (11); schedule appointments, change appointments, call nurses/insurance, make a list of medications (1); assist with billing and insurance processing and problems (1); time for labs (1); help with prescription fill/refill (1).

These results indicate to us that the biggest need patients have that digital tools can help mitigate is the administrative functions of managing their care. This is corroborated by repeated surveys and focus groups about the number of hours patients spend managing their care, often likening it to a full-time job. Specifically, patients consistently report that care management is too fragmented, with too many portals and not enough interoperability across portals, apps, and Electronic Health Record (EHR) systems. Ideally patients would have a one-stop-shop that allows them to access all their health data in one place, easily switching between provider types; and would have full transparency into the status of prior authorization requests, appeals, and any other benefit verification processes.

Just as (if not more) importantly, these streamlined functions would be actionable. For administrative functions, patients could facilitate interactions with any given stakeholder, rather than making phone calls and ping-ponging between providers, pharmacists, and health plans for potentially hours, as is often the case now. For clinical functions, patients have express a need for integration of clinical data in one place that can facilitate "quarterbacking," allowing one provider to see all the clinical notes related to that patient across providers in order to make more informed and holistic treatment decisions. People with chronic diseases like arthritis often rely on multiple providers to manage their disease, and there is currently limited (if any) capability of this nature, putting the patient in the position to have to interpret clinical notes from other providers to the best of their ability. For apps that patients use to manage and track symptoms, digital tools would have bi-directional functionality such that patients can receive guidance and recommendations for next steps based on their inputs.

Another critical consideration is the role of the caregiver. People with arthritis often rely on caregivers to help manage their care, so it is critically important to address caregiver

needs and ensure they can easily access and benefit from these digital tools if authorized by the patient.

Another wish list item we often hear from patients concerns redundancies in providing demographic data. We often hear complaints of how often patients are repeating the same demographic data, even within the same provider practice. Ideally, they could input basic demographic and health status data once and enable that data to populate across all their providers' electronic systems. While there have been advances in the check-in process many provider practices now utilize, additional advancement and utilization of seamless digital check-in processes can also lessen the administrative burden of provider visits.

Finally, at the point of care having access to real-time benefit tools (RTBT) to understand the costs and utilization management requirements for any given treatment option could be game-changing for patients. While aspects of RTBT does currently exist, there is much room for improvement and broader availability to be truly patient-centered.

PC–2 Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

- **a. If so, what are some examples of benefits it has provided?**
- **b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?**
- **c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?**

In our survey, 25 respondents reported they can see all their health information in one place, with 10 respondents indicating they could not. Lab tests and imaging results frequently came up as the type of data patients want easier access to; often those results are difficult to access. One patient we spoke with who has to have blood work routinely said it has been incredibly helpful having her results logged in one place and being able to see the trend lines of her results longitudinally. It has helped her to flag when she knows there is an anomaly and essentially has made her a more informed patient partner in her own care. These are the types of benefits that could greatly improve patient-centered care.

PC–3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

In our survey, 27 respondents reported using digital health apps or websites to help manage their care, versus 10 respondents who do not. There are many digital health tools

on the market that can facilitate a wide range of services from administrative care management to symptom tracking, and remote therapeutic monitoring (RTM). We performed an online search of arthritis-specific apps to understand the user experience navigating the sheer volume of information and tools available. What we found is a multitude of apps available for those living with arthritis to help track and manage their systems. Most apps available provide some level of symptom, medication, and lifestyle tracking. Some apps are associated with specific hospital systems which allow patients to connect with providers and specialists, but the number of hospitals is very limited. Currently, there is one app in the arthritis space approved by the FDA, PatientSport which evolved from ArthritisPower. When searching terms related to "arthritis" and "arthritis management," apps not specifically for arthritis come up in the search. Most of these apps function as logs for symptoms, mood, medication, and treatment. There are a handful of apps that feature AI integration and features, which will only continue to grow in the coming years.

Some apps serve to connect patients with other patients or translate health information into easy-to-read reports. These sorts of functions can be enormously helpful and valuable to patients if utilized correctly. Some symptom and wellness tracking apps are covered by various insurance companies, and there is at least one app by a pharmaceutical manufacturer of an arthritis biologic therapy that offers support for the self-administration and management of biologics. Tailored tools like this that can be at a patients fingertips any time can increase comfort levels and adherence of medication, as newly diagnosed patients in particular can experience significant anxiety when learning to self-inject their medication.

However, it can be difficult to navigate the credibility and reliability of many health metrics and tracking apps, and aggressive or deceptive marketing practices coupled with paid algorithm optimization can lure patients into utilizing products that are not FDA-approved or validated. Per our previous comments, many of these apps are not HIPPA covered entities and therefore CMS and other stakeholders should prioritize patient education around privacy and security of their personal health information (PHI). The App Store does have an "App Privacy" section that gives an overview of data privacy protections, but it may not be intuitive or even know by users to utilize this function.

Questions our search left us with that we encourage CMS to consider include:

1. How does search engine optimization and app store optimization alongside paid placement skew what is presented to patients when searching for apps to help manage their condition?
2. How did AI-enabled tools enhance or hinder the app experience? Does AI integration create a higher level of trust with users or a level of distrust?
3. Do rankings and ratings impact a patient's decision on whether to download and use an app or not?

PC–4. What features are missing from apps you use or that you are aware of today?
a. What apps should exist but do not yet? Why do you believe they do not exist yet?
b. What set of workflows do you believe CMS is uniquely positioned to offer?

The Arthritis Foundation believes CMS is uniquely positioned to set standards and requirements for the exchange of health data, and to build upon rule 0057 regarding patient access APIs, other APIs, and prior authorization transparency requirements. We supported and applauded CMS for finalizing 0057 and believe it is crucially important that CMS develop and finalize a rule adding prescription drugs to the prior authorization requirements.

CMS can also work within its purview to facilitate further interoperability and data exchange across sites of care, including physician offices, hospitals and facilities, and home. The ultimate goal is that health care is following the patient throughout their journey, regardless of setting, rather than the current flawed paradigm where the patient is often following the health care.

PC–5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

a. **What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)?**
b. **What criteria should be used if there is a review process?**
c. **What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?**
d. **What changes would enable timely access to high quality CMS and provider generated data on patients?**

CMS should work closely with the FDA on the review of efficacy, quality, and impact of digital health products on health outcomes. For example, CMS can support FDA post-market surveillance by collecting and analyzing data around uptake and utilization of digital health products from claims data and available EHR data. To the extent possible, CMS can work directly with health plans to understand the benefits and challenges associated with using digital health products from the patient perspective, and can evaluate uptake based on outreach and education to patients about the availability of digital health tools.

Another critical role CMS can play is in setting clear definitions for digital tools that are FDA-approved and vetted versus those that are not; and prioritizing clear disclosure to

patients about the difference. As mentioned previously, the market is currently flooded with digital tools under the umbrella of "health and wellness" but we know there is a vast range of efficacy and reliability.

Relatedly, patients often express concern about data privacy and security in discussions about digital storage of their health data. CMS should prioritize patient education and disclosure on the different standards for HIPPA covered entity and apps that are not covered by HIPPA. It is highly likely that most patients do not understand the different circumstances in which their PHI are protected, and CMS can play a crucial role in providing guidance to beneficiaries as they select digital health tools.

Finally, as the end users of these products, CMS should include patient reviewers in its review processes. It is our policy as a patient advocacy organization to always include patient reviewers in our work products and the input they provide is invaluable. Organizations like ours are ready and willing to assist in patient engagement processes and we urge CMS to utilize us as a resource.

PC–6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Our survey data indicate there are multiple factors that hinder broader use of digital health products. For some patients it's a lack of belief that they need to use digital health products. For others, particularly rural and/or elderly beneficiaries, it's lack of access to computers or wifi. For many, it's the difficulty of keeping track of passwords. We heard over and over in our survey that one of the largest obstacles keeping patients from more consistently using apps and portals is remembering the many passwords required for all the portals and apps across their providers and insurance companies. Addressing these obstacles will be necessary to increase uptake of digital health products. Ensuring they are free or low-cost is important for accessibility, and having patient-tested, intuitive interfaces will help with ease of use. Again, this is an area in which the Arthritis Foundation would welcome the opportunity to be of assistance.

PC–7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

CMS should utilize CMMI to test and evaluate methods of collecting real-world data on digital health products. CMS can also leverage claims data and EHR data where possible to evaluate cost and health outcomes. Patient reported outcomes data and patient organization survey data can help supplement other data sources. Data from AI-enabled technologies and RTM technologies can also be impactful for building real-time, longitudinal data points on specific pathways of care like physical therapy.

2. Data Access and Integration

PC–9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers? a. What difficulties are there in accessing or utilizing these data sources today? b. What suggestions do you have to improve the Blue Button 2.0 API experience? c. Is there non-CMS data that should be included in the API?

From the patient perspective there is a lack of knowledge about APIs, what they are, and how they can be utilized to improve care coordination. There is an important role for CMS, health plans, SHIP counselors, and providers to educate beneficiaries about APIs and assist beneficiaries in how to select and utilize one.

PC–12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

> **a. Examples may include the following: (1) Binding cost estimates for predefined periods. (2) Viewing provider schedule availability. (3) Using third-party apps for appointment management. (4) Accessing patient facing quality metrics. (5) Finding the right provider for specific healthcare needs.**
> **b. What use cases are possible today?**
> **c. What should be possible in the near future?**
> **d. What would be very valuable but may be very hard to achieve?**

A theme that emerged from our survey is that simple capabilities like managing appointments would make a big difference for chronic disease patients. Being able to schedule appointments through digital tools came up often; this could streamline care significantly given that people with chronic disease need regular access to care, often with multiple providers. Other factors that emerged from the survey include provider availability and accurate information about in-network providers.

3. Information Blocking and Digital Identity

PC–13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

The first and most crucial thing CMS can do is educate patients about the information blocking rule. In our survey, only 8 of 37 respondents had heard of the Information Blocking Rule. When we asked what would help more people to know about it, the most

frequent answer was it should be visible and easy to find when logging into their insurance portal. One respondent indicated it was hard to find and they only learned of it because their mother-in-law is a physician. Once a patient learns about the rule, they need to then know how to report complaints. A couple of respondents indicated they didn't know what good knowing about the rule would be if they don't know what to do with it, and/or that they wouldn't even know where to start in terms of finding out how to report complaints.

Aside from basic education about where and how to report a complaint, CMS must clearly explain to patients the value of doing so. For some, they need to know there is a benefit to reporting a complaint and that it will result in meaningful action. For others, they are busy managing their chronic illness (often more than 1) and the thought of "using their limited spoons to report a doctor is often overwhelming." CMS should indicate how complaints support enforcement of the rule and how CMS utilizes the data from complaints to improve patient rights.

PC–14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800–63–3 IAL2/AAL2 credentialing service providers (CSP)): a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials? b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used? c. What are the potential downsides? d. How would encouraging the use of CSPs improve access to health information? e. What role should CMS/payers, providers, and app developers have in driving adoption? f. How can CMS encourage patients to get digital identity credentials?

In our survey, 23 of 37 respondents had used some form of digital identity tool to access health information, including three-quarters of Medicare respondents. For those that don't, the most common reasons they gave for not using them were not knowing about the options, and not having a need for such tools, with one respondent voicing privacy concerns.

Our patient feedback highlights three main areas of focus for CMS in driving adoption of digital identify credentials:

1. Basic knowledge of the tools: CMS can better inform and advertise such tools through payers, provider offices, and its own resources on Medicare.gov.
2. Addressing the benefits: with many respondents indicating they don't see a need for such credentialing tools, CMS should clearly lay out how this technology can benefit patients in their care management.
3. Privacy concerns: CMS should prioritize ensuring credentialing standards safeguard patient data and should clearly disclose to patients how their data is utilized.

F. Value Based Care Organizations

1. Digital Health Adoption

VB–2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

There is great opportunity to enhance value-based care through technology. AI-enabled technologies are already helping to streamline the administrative side of health care management, and there is great promise in clinical applications. In a 2024 panel we conducted specific to AI, a pediatric rheumatologist likened treating patients to editing a movie: you only get to see every few scenes (i.e. how the patient is doing at the provider visit) but the majority of the action is happening in the scenes in between (i.e. daily patient symptom management). AI-enabled technology could bridge this gap through functions like remote-therapeutic monitoring and symptom-tracking apps that can connect directly to their providers' EHRs.

A promising study soon-to-be published in the Archives of Rehabilitation Research and Clinical Translation show that RTM in combination with physical therapy for musculoskeletal (MSK) conditions like osteoarthritis reduced pain and increased adherence to therapy. It showed that a greater proportion of patients using RTM and physical therapy versus physical therapy alone experienced higher functional status scores and higher engagements scores. This can possibly be attributed to the access to education, support, and provider engagement between clinical visits. Overall, this type of technology can increase adherence and improve patient outcomes, thereby positively impacting long-term health.

Similarly, a Peterson Health Technology Institute study assessing virtual solutions to MSK exercise management showed that app-based virtual exercise solutions are associated with improved pain scores relative to no in-person PT; and PT-guided therapy solutions and RTM augmented therapy solutions improved pain scores and adherence even further. They concluded that virtual therapy options can close access gaps, particularly for the elderly and rural populations.[1]

These types of health care interventions are ripe for inclusion in value-based care models, as they allow for patient-centered, evidence-based care management outside the physical provider office. While some of these products are covered by Medicare, reimbursement models remain a hurdle, particularly for scaling to other insurance types. Further, there is currently no standard reimbursement pathway for FDA-approved AI technologies, and

[1] https://phti.org/wp-content/uploads/sites/3/2024/06/PHTI-Virtual-MSK-Solutions-Executive-Summary-1.pdf

few of them are reimbursed by Medicare. We have supported legislation such as the Health Tech Investment Act that would create such a pathway, but encourage CMS to identify ways to mitigate this barrier as well. We also encourage CMS to work with a broad array of stakeholders to identify the most efficient and cost-effective ways of incorporating AI-enabled technology into APMs and other care models.

That said, there is a significant amount of apprehension and mistrust in AI, so CMS should prioritize accuracy, bias mitigation, and data privacy and security in its AI-related workstreams.

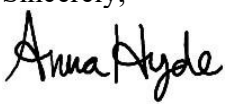
We conducted an advocate survey in 2024 with an AI-specific question block; relevant highlights include:

- Half of all respondents indicated having little or no trust in AI technologies for making accurate and beneficial decisions in arthritis care.
- Only 17% of respondents have embraced AI for arthritis care with wearable devices topping the list for those that have used it.
- While nearly half of all those using AI for healthcare indicate a high comfort level with it, the results vary by age group with seniors having the lowest comfort level.

Interestingly, it was not just older age cohorts that had low levels of comfort in using AI-enabled digital tools. In patient panels we have found that younger generations, specifically those now entering their 20's, also have apprehension around having their health care digitally tracked. We hope these data points show the value in engaging patients in policy decisions and implementation of digital health care tools, and urge CMS to fully investigate these types of user preferences when designing health technology policies.

Thank you again for the opportunity to comment and we look forward to working with you to improve the digital health ecosystem for patients living with chronic disease. Should you have any questions or if we can be of assistance, please contact me at ahyde@arthritis.org or 202-843-0105.

Sincerely,



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